Ashley

The Wayne State grid and Claude Code, from one wizard. Manual, version 1.0.0

2026-09-28

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> Ashley is a VS Code extension for medical students in OpenSpineConsortium projects. It sets up the laptop for the Wayne State HPC grid, keeps a Claude Code worker running there on the student's own subscription, and lets the Claude Code extension on the laptop hand work to it. This manual is the pitch, the first hour, how it works, what you can and cannot ask, installation, and troubleshooting. The design notes and the research behind them are in the repository (`docs/ARCHITECTURE.md`, `docs/RESEARCH.md`).

Why

For Wayne State medical students joining an OpenSpineConsortium project.

You know things no computer scientist on the grid knows. You know why a sacral slope of 9° is implausible, what a transitional vertebra does to a surgeon's count, which patients get the second operation. What you may not have is the year it takes to turn that knowledge into a measurement, a job on 802 CT scans, a figure and a paragraph in a paper. Ashley is built to remove that year.

What Wayne State already gave you

The university runs a grid with more than four hundred compute nodes, NVIDIA H200, L40S and A30 GPUs, hundreds of CPU cores you can hold at once, four terabytes of personal storage and twenty more on the lab's shared volume, and an internet connection from every compute node. Every student with an AccessID may apply for an account. Almost none do, because the entry fee is a week of learning SSH, Slurm, containers and shell quoting, and one misstep runs your job on the login node and gets it killed.

Ashley pays that fee for you. One wizard: your AccessID, one password and one Authenticator code, your GitHub and Hugging Face tokens, and a click in the browser that lets Claude work on the grid as you. From then on the grid never asks you for anything.

What you get

A Claude that lives where the data is. Ashley starts a Slurm job, on a CPU node or one with a GPU, that runs Claude Code inside a container in your grid space, with your own subscription. You talk to it from the Claude Code panel in VS Code, or from a prompt box, or from a phone over ssh if you like. You say *"in my project folder, measure the bi-iliac width on the ten development cases and show me the numbers with units"* and it

reads the manifest, writes the script, runs it, looks at the output, fixes what is wrong and answers. Close the laptop. It is still running.

A conversation that does not forget. Sessions are named. *"In session `thesis`, what did we decide about the exclusion list?"* works next week, from a different node.

Your files, in your editor. One command opens your grid home in VS Code on the compute node where your job runs: Explorer, editor, terminal, and a click on a PNG shows the figure. No X server, no copying files back and forth.

Two Claudes that work together. The Claude Code extension on your laptop gets a handful of tools for the grid. It can plan, and hand the heavy step to the grid worker, and bring the answer back into the panel. You can say *"check the grid, run the pelvic width demo on ten records, and tell me whether the sex difference matches the published value"* in one sentence.

Honesty about limits. When your subscription window is used up, the worker says so, tells you when it resets, and waits. It never uses anyone else's account, never types a token on a command line, and never runs anything heavy on the login node.

What becomes possible

- A first-year student with an idea on Monday has ten measured cases by Tuesday and a pre-registered plan by Friday, because the parts that used to take a month (environment, data path, job script, first plot) now take an afternoon.
- A lead runs three projects at once: each has its own worker, its own sessions, its own journal written by the worker at the end of every task.
- A literature question becomes a scoping review pipeline: the worker searches PubMed and OpenAlex with your criteria, logs every inclusion and exclusion, and drafts the PRISMA table, overnight, on a node you are not paying for.
- A model that needs a GPU trains on an H200 while you are in clinic; the loss curve is a PNG in your grid folder when you come back.

What it asks of you

Follow the recipe. Write the question before the number. Freeze the pre-registration. Let the worker run scripts and jobs, and read what it produced yourself: render the mask, check the spread against the literature, ask what could make a plausible number wrong. Ashley makes the machine easy; the judgement is still the point, and it is yours.

Getting started

Install the extension, run **Ashley: Set up my research environment**, and start a worker. The manual walks through every step with screenshots of what you will see. Your lead can add you to a project card on the board the same week.

Getting started

Before you open VS Code

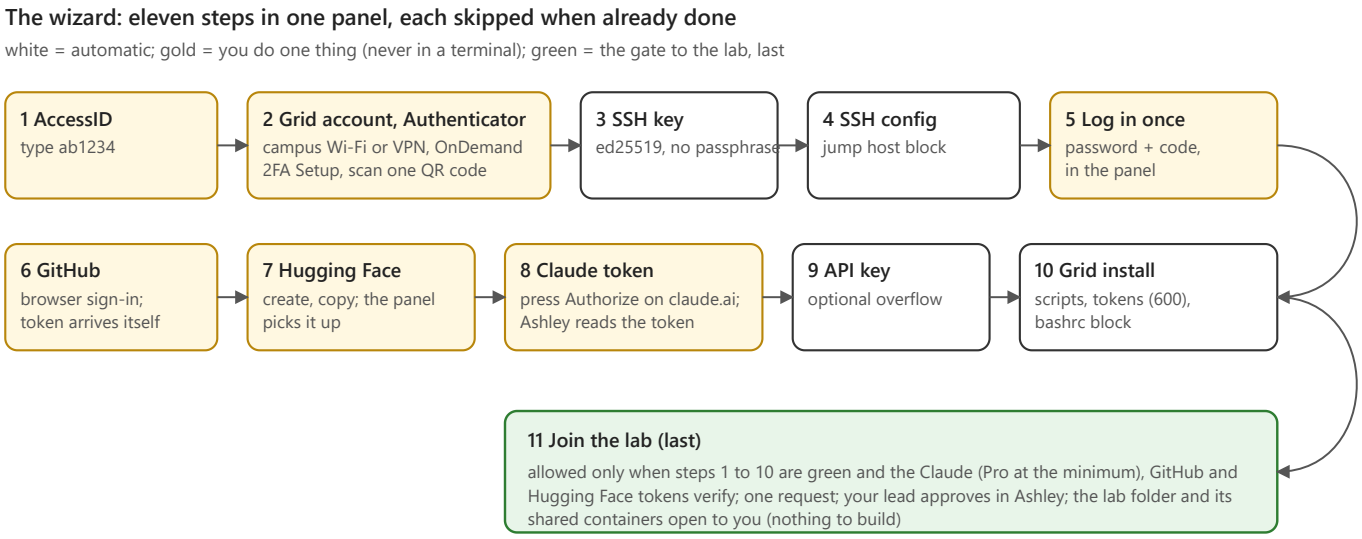
1. **A grid account** (two business days) and **Google Authenticator** for it. The wizard's second step walks you through both: it opens the request page, tells you whether the WSU VPN is on, opens OnDemand's 2FA Setup and the C&IT guide. If you already have them, you say so and move on.
2. **A Claude Pro or Max subscription**, and the Claude Code extension signed in.
3. **GitHub and Hugging Face accounts.** Create them with your WSU email if you do not have them; the lab adds you to its GitHub organisation by that address.

Install

VS Code, then the extensions **Claude Code**, **Remote - SSH**, and `ashley-wsu-1.0.0.vsix` (Extensions view, `...`, *Install from VSIX*). Ashley offers its wizard on first start; or press Ctrl/Cmd+Shift+P and type **Ashley: Set up my research environment**.

The setup panel

Everything happens in one panel: the eleven steps on the left with their state, the current question on the right, a progress bar on top, the log one click away. Finished steps are skipped, so the wizard can be rerun at any time. Nothing is typed into a terminal.



The nine wizard steps

Step	What you see in the panel	What you do
1 AccessID	a text field	type it (`ab1234`)
2 Grid account and Authenticator	a menu: the account request page, OnDemand (the panel says whether the WSU VPN is on), the C&IT guide, the Authenticator app for iPhone or Android, and *continue*	apply for the account if you have none; connect the VPN; on OnDemand open Interactive Apps, 2FA Setup, Launch, Launch 2FA Setup, and scan the QR code with Google Authenticator; then *continue*
3 SSH key	a spinner for a second	nothing
4 SSH configuration	a spinner for a second	nothing
5 Log in to the grid once	a password field, then a code field	type your grid password, then the six-digit Authenticator code. Once. Ashley installs its key and the grid never asks again.
6 GitHub	your browser opens on GitHub's sign-in	sign in and approve; the token arrives in the panel by itself
7 Hugging Face	a button *Open the page in my browser*; the token page opens pre-filled (name `wsu-grid`, type Write)	press *Create token*, then *Copy*; the panel notices the copied token and offers *Use it*
8 Claude token	a terminal titled *claude setup-token* opens below; the browser asks you to approve	approve, copy the printed `sk-ant-oat...` token; the panel notices it and offers *Use it*
9 API key	a field	leave it empty and press *Skip* unless your lead gave you a key
10 Install on the grid	a progress line; "installed; the lab's container comes with lab access" (or "adopted the lab's shared container" once you have access)	nothing
11 Join the lab	a note field, only once every step above is green and your tokens verify	write your project and lead; one request goes to your lead, who approves it in Ashley. From then on your workers use the lab's shared container, nothing to build

The last step is refused until everything above works, on purpose: the lab admits working environments. **Ashley: Lab access to the shared library** shows the state of your request afterwards (pending, approved, or denied with your lead's message).

If a token cannot be checked from your laptop's network (some campus Wi-Fi blocks it), the panel keeps it and the grid verifies it a moment later; a token the service actually rejects is refused with a reason.

The wizard writes only: a key pair and one block in `~/.ssh/config` on the laptop; tokens in the OS keychain; on the grid `~/.ashley/`, a block in `~/.bashrc`, and a container in your storage.

Your first worker and your first question

1. **Ashley: Start a Claude worker on the grid** (after your lead approved step 11): name `default`, CPU node, 24 hours, and which lab container (the default is right for most work). A Slurm job is submitted. When it starts, the Ashley view shows `Worker default <node> idle`.
2. **Ashley: Ask the grid Claude**: type *"Create a folder hello in my home with a README that says which node you are on and what Python version you have."* The answer streams into an output channel; tool calls appear as italics.
3. **Ashley: Open my grid files on a compute node**: a new VS Code window opens on the node; you see `hello/README.md` in the Explorer. The window shows a banner that the OS is unsupported: expected on the grid's CentOS 7 (Wayne State supplies the compatibility libraries); dismiss it.

Talking to it from Claude Code

Ashley: Connect the Claude Code extension to the grid registers the bridge (once). Then, in the Claude Code panel: *"Check the grid status and tell me whether my worker is idle."* The local Claude calls `ashley_status`. For work: *"Ask the grid Claude, in session `demo`, to run `examples/pelvic_width_dimorphism/measure_pelvic_width.py` on ten records and report the mean."*

Starting over, or showing a friend

Ashley: Reset this laptop (emulate a fresh install) forgets, as you choose, the tokens, the key and config block, the settings, the Claude Code registration, and Ashley on the grid. After a reload, VS Code looks the way it does to a brand-new student, and the wizard runs from the start. To emulate a brand-new VS Code as well, start it with its own profile folders: `code --user-data-dir %TEMP%\fresh-vscode --extensions-dir %TEMP%\fresh-ext` (macOS/Linux: `/tmp/...`), install the three extensions there, and run the wizard.

Every day after that

- Start a worker when you sit down (or keep one running for a day); it costs nothing.
- Name your sessions after the project; a session remembers its folder.

- Let the worker write scripts and Slurm job files; you read them; you say “submit”.
- Pull small results back with git or `rsync`; open PNGs on the node with Remote-SSH.
- If something is off: **Ashley: Check everything (doctor)**. Its notification names the step; the details are in the Ashley output.

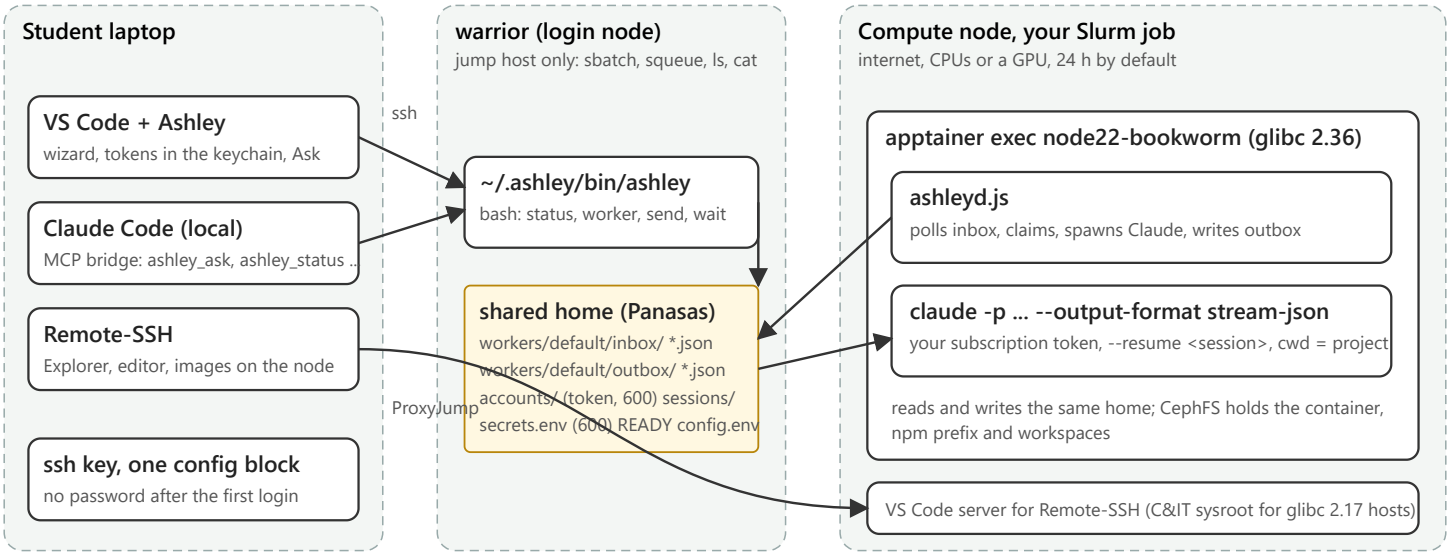
How it works

Ashley is three small things that agree on one file format.

Piece	Runs on	Language	Job
VS Code extension (`extension/`)	the student's laptop, inside VS Code	TypeScript	wizard, secrets, Remote-SSH, the Ashley view, "Ask", MCP registration
MCP server (`extension/dist/mcp-server.js`)	the laptop, spawned by Claude Code	TypeScript (bundled)	gives the local Claude the `ashley_*` tools
Grid CLI + daemon (`grid/`)	the grid: `ashley` on the login node, `ashleyd.js` inside a container on a compute node	bash, Node	install, workers, the request queue, Claude Code itself

Everything the laptop does on the grid is `ssh wsu-grid ~/.ashley/bin/ashley <verb>`. There is no port, no tunnel and no server on the laptop; nothing on the login node beyond `sbatch`, `squeue`, `cat`, `ls` and a half-second sleep loop.

Topology



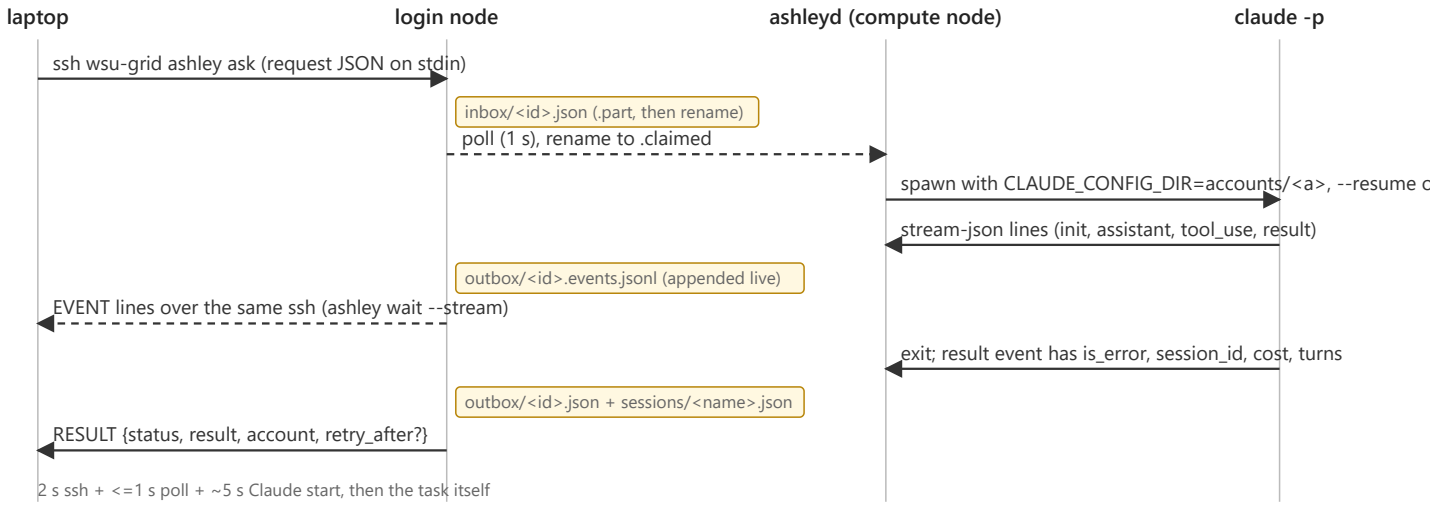
The three places and what runs in each; every laptop action is one ssh to the login node.

```
laptop                          warrior (login node)              compute node (Slurm job)
──────                          ────────────────────              ────────────────────────

VS Code — Ashley ext ──ssh──▶ ~/.ashley/bin/ashley ──files──▶  apptainer exec node22-bookworm
Claude Code — MCP ─┘           (bash: install, status,          └─ node ashleyd.js
Remote-SSH ──────────ProxyJump──────────────────────────────▶     └─ claude -p ... (2.1.283)
                               sbatch / squeue                     watches  workers/<w>/inbox
                                                                   writes   workers/<w>/outbox
                               shared home: ~/.ashley/{accounts,sessions,workers}
                               CephFS ROOT: containers/, npm/, workspaces/
```

The login node is a jump host. VS Code Remote-SSH connects to `voh3.nas.grid.wayne.edu` (any node where the student holds a job) through `ProxyJump wsu-grid`; C&IT forbids the VS Code server on `warrior` and kills it there. Compute nodes have internet, so Claude Code, `npm`, Hugging Face and Zenodo all work from inside the job.

The request queue



One request, from the laptop's ssh to the RESULT line.

A request is a JSON file. Its life:

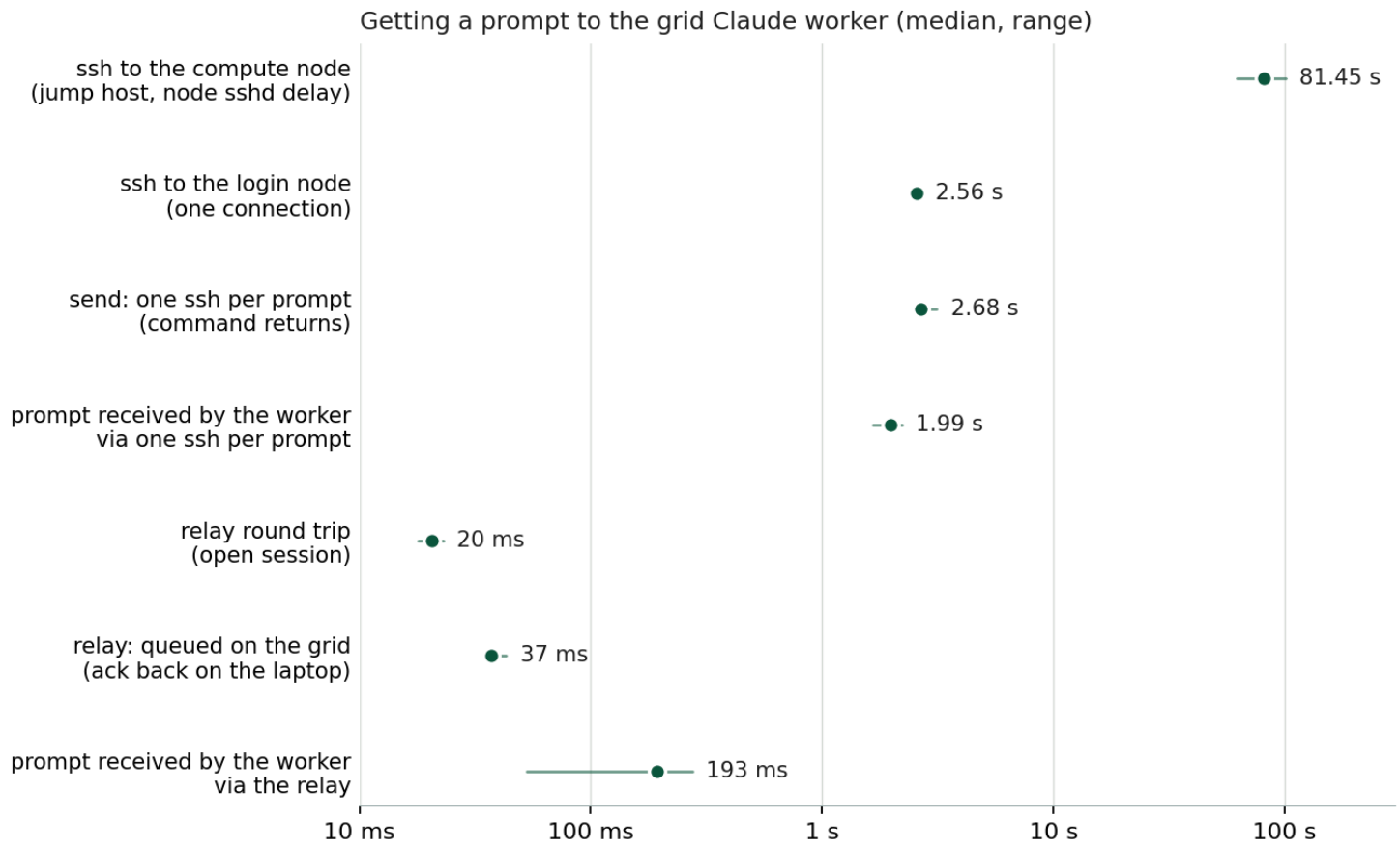
1. The laptop writes `~/.ashley/workers/<w>/inbox/<id>.json` through `ashley send` (stdin, `.part` then rename, so the daemon never reads a half-written file).
2. `ashleyd.js` polls the inbox every second, claims the file by renaming it to `<id>.json.claimed` (an atomic operation on the shared filesystem; two daemons cannot both win).
3. It picks a credential (below), resolves the session, and spawns `claude -p <prompt> --output-format stream-json --verbose --permission-mode <mode> --max-turns N [--resume <sid> | --session-id <uuid>]` in the session's working directory, with `CLAUDE_CONFIG_DIR=~/.ashley/accounts/<account>`.

4. Every stream-json line is appended to `outbox/<id>.events.jsonl` as it arrives. `ashley wait --stream` tails that file to the laptop as `EVENT {…}` lines, so VS Code shows tool calls live.
5. On exit, the daemon writes `outbox/<id>.json` (`.part` then rename): status, result text, session id, account, cost, turns, duration, node, job. `ashley wait` prints it as `RESULT {…}`.

How a prompt travels, and how fast

The laptop keeps **one ssh session** to the login node open, running `ashley relay`. A prompt is one line on that session's stdin; the relay writes the inbox file and answers `QUEUED`; the daemon polls the inbox every 250 ms; results and events come back on the same session as `RESULT` / `EVENT` lines. Measured on 2026-09-28 (Windows laptop, skew-corrected; `docs/research/`):

Path	Median	Range
one ssh connection to the login node	2.6 s	2.4 to 2.7 s
prompt received by the worker, one ssh per prompt (the first design)	2.0 s	1.6 to 2.2 s
relay round trip	20 ms	18 to 23 ms
relay: queued on the grid, ack on the laptop	37 ms	35 to 43 ms
prompt received by the worker via the relay	**0.19 s**	0.05 to 0.28 s
ssh straight to the compute node (for comparison)	about 25 s	the node's sshd delay



Windows 11 laptop, Wayne State grid, 2026-09-28. Receipt = worker's claim time minus the laptop's send time.

Latency of the prompt paths

So the relay is roughly ten times faster to receipt and the remaining 0.1 to 0.3 s is the daemon's poll interval. It is "UDP-like" in the sense that matters (no per-message handshake, fire-and-forget `send` with an ack in 40 ms) while staying reliable: the file is the message, and a dropped session loses nothing that was already queued. If the session drops, the next call reconnects; every call also falls back to a fresh ssh. True UDP to the node is impossible (the laptop cannot reach compute nodes) and unnecessary. Claude Code's own start-up (about 5 s cold) now dominates the time to a first token; a warm `--input-format stream-json` process per session is the next lever if that ever matters.

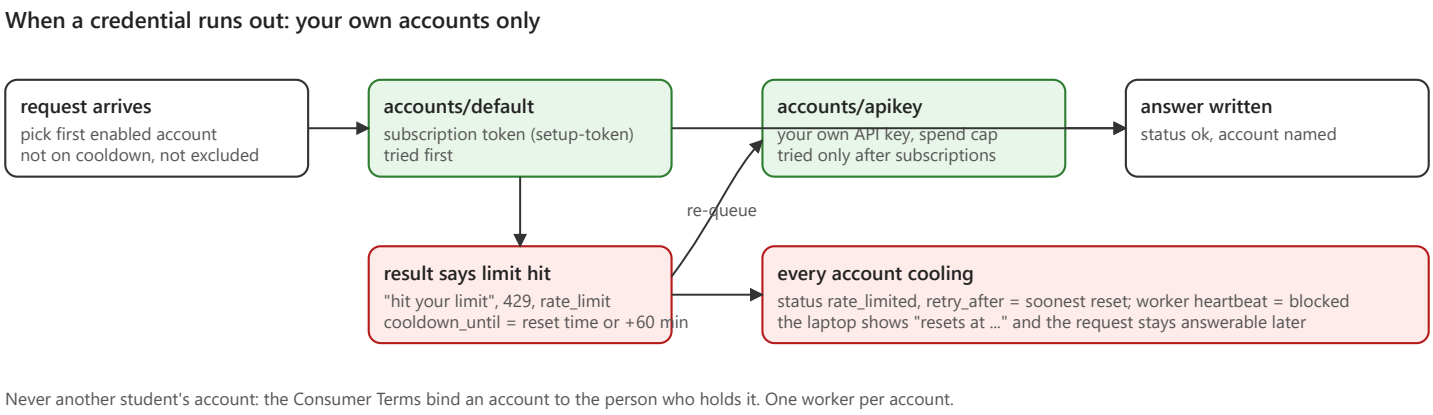
A queued request survives a daemon restart, a requeued job or a closed laptop: the daemon moves any `.claimed` file back to the inbox when it starts. An interactive ssh to the login node or the node (Remote-SSH, `Ashley: Open a terminal on the grid login node`) remains available; it is simply not the path prompts take.

Why not a socket or a TCP port: the laptop can only reach the login node; the daemon runs on whichever node Slurm chose; the shared home is the one thing both see. Files are also what a student can debug with `ls` and `cat`.

Sessions

`~/.ashley/sessions/<name>.json` maps a name the student chose (“thesis”, “figure-2”) to a Claude Code session id, the working directory it is bound to and the account that owns it. The first turn of a session passes `--session-id <uuid>` so the id is known even if the run dies; later turns pass `--resume <uuid>`. Claude Code keeps the transcript under `<CLAUDE_CONFIG_DIR>/projects/<encoded cwd>/<uuid>.jsonl`, which is why a session stays bound to its directory.

Credentials



Rotation among a student’s own credentials; a limit becomes a cooldown, not a failure.

Each `~/.ashley/accounts/<name>/` is both a credential and a `CLAUDE_CONFIG_DIR`, so accounts never share sessions, settings or caches:

- `token`: a long-lived subscription token from `claude setup-token` (the normal case). Exported to Claude Code as `CLAUDE_CODE_OAUTH_TOKEN`.
- `apikey`: an Anthropic API key (pay-per-use overflow). Exported as `ANTHROPIC_API_KEY`.
- `.credentials.json`: a copied interactive login. Works until its refresh token dies; not recommended (the probe on 2026-09-28 showed a copy rejected after eight idle days).

Order of use: subscriptions before API keys, `default` first. When a run ends with an error that matches the limit patterns in `protocol.ts` (“hit your limit”, “rate_limit”, “429”, …), the account gets `state.json` with `cooldown_until` (the reset time Claude printed, else one hour), the request is re-queued for the next enabled account, and if none is left the result is written with `status: rate_limited` and `retry_after`. The worker’s heartbeat reports `blocked` with `blocked_until` so the laptop can say “your window resets at 3 pm” instead of failing.

One person, one token. Pooling several people’s subscriptions into one worker would breach the Consumer Terms (each account is for the person who holds it); the rotation exists for a student’s own second credential, typically an API key with a spending cap.

Secrets: where they are and where they are not

Secret	Laptop	Grid	Never
GitHub, Hugging Face tokens	VS Code SecretStorage (OS keychain); injected into integrated terminals	`~/.ashley/secrets.env` (mode 600), read by the daemon into the worker's environment	job logs, `sbatch --export`, command lines
Claude subscription token	SecretStorage	`~/.ashley/accounts/default/token` (600)	laptop terminals
Anthropic API key	SecretStorage	`~/.ashley/accounts/apikey/apikey` (600)	laptop terminals (it would silently switch the local Claude Code to pay-per-use)
Grid password / OTP	nowhere	nowhere	typed once, in a terminal, to install the SSH key

Files travel over ssh **stdin** (`cat > file` with `umask 077`), never as arguments.

The container

`bootstrap.sbatch` (one job per user, 7 minutes measured) builds an unprivileged Apptainer **sandbox directory** from `docker://node:22-bookworm` on the student's CephFS space and installs `@anthropic-ai/claude-code@2.1.283` into a persistent npm prefix there. A sandbox rather than a `.sif`: SIF-to-sandbox conversion at exec time is what failed on this grid before, and several nodes lack `unsquashfs`. The host is CentOS 7 (glibc 2.17); Claude Code needs Node 22, which needs a newer glibc, hence the container (glibc 2.36 inside). Runtime order tried: `module load apptainer/1.2.2` (worked on `asx1`), the conda `singularity` 3.8.6, `module load singularity/3.5.2`, `/usr/bin/apptainer`. `~/.local/bin/singularity` is skipped on purpose.

The worker job (`worker.sbatch.tpl`) execs the daemon inside the sandbox with `$HOME`, the CephFS tree and a `/dev/shm` scratch bound to `/tmp` (`/tmp` on nodes is swept by a cleaner while jobs run). `--signal=B:TERM@120` gives the daemon two minutes to finish a turn at the wall limit; `--dependency=singleton` keeps one job per worker name; on the `requeue` QoS the job is `--requeue --open-mode=append` because the daemon is re-entrant.

The shared container library

Containers are read-only at run time (every account's sessions, settings and caches live in its own home), so one copy serves the whole lab. `/rs/rs_grp_oschome/shared/ashley/` (group `oschome`, mode 2775) holds `library.json` and one sandbox per entry:

```
{"version": 1, "default": "claude-node22",
 "containers": {"claude-node22": {"sandbox": ".../containers/claude-node22", "claude_bin":
        ".../npm-claude-node22/bin/claude",
   "runtime": "/wsu/el7/gnu7/apptainer/1.2.2/bin/apptainer", "image": "docker://node:22-
        bookworm",
   "claude_version": "2.1.283 (Claude Code)", "gpu": false, "built": "...", "built_by":
        "go2432", "purpose": "general ..."}}}
```


- `ashley install` adopts the library's default when it is readable (`ashley adopt <name>` writes `config.env` and `READY` in a second; nothing is built, nothing is stored in the student's space). `ashley install --build-own` keeps the old per-account build.
- `ashley worker start --container <name>` runs any entry; the worker's job script carries the sandbox and binds the library root. A student can run a CPU worker on the general container and a GPU worker on a CUDA one at the same time.
- Leads publish entries with `ashley library publish --name <name> --image docker://<image> [--gpu] [--purpose "..."]`: a job builds the sandbox, installs the pinned Claude Code, merges the entry into `library.json` and makes the tree group-readable. Entries are pinned by image and Claude Code version, so every student runs the same bits; an update is a new entry.
- **Lab access is one folder, self-service for the lab admin, no HPC ticket.** The lab folder is `/rs/rs_grp_oschome/shared` (the containers now, anything the lab adds later). Only its `access/` sub-folder is open to everyone: `inbox/` (1733, a sticky drop-box: a student creates their own request file and cannot list or touch others'), `status/` (711: each student reads `status/<own id>.json`, nobody can list), `requests/` (700, the admin's archive). Approving runs `setfacl` (CephFS honours POSIX ACLs and the admin owns the volume): `r-x` plus a default ACL on `shared/`, and `r-X` recursively with default ACLs on everything under it except `access/`, so folders the admin adds later are covered too. Deny, remove and reset take it away; the student never joins the `oschome` unix group. The admin's own tree (`/rs/rs_grp_oschome/<admin>`, 2750) stays private.
- **The request is the wizard's last step (`labAccess`) and is refused until the environment works the way the lab works:** every earlier step done; the Claude Code and Remote-SSH extensions and the bridge present; and, checked on the grid by `ashley access requirements`, a Claude subscription token that the API accepts (one `GET /v1/models` with the OAuth token: 200 = valid, 401 = invalid or revoked; `claude setup-token` only issues tokens to Pro/Max and above), and GitHub and Hugging Face tokens that verify. The shared container is **not** a requirement: it is what access gives. Until approval `ashley install` does not build a private copy (it marks `awaiting_access`); `ashley install --build-own` is the escape hatch. After approval, the next `worker start` or wizard run adopts the lab container in a second.
- Integrity of the drop-box: the admin's ingest (run as the admin) accepts a request only if it is a regular file, not a symlink, under 4 KB, and owned by the user it names; anything else is moved to the archive as rejected. One request per student until the admin resets

it. Token checks pass tokens to `curl` on stdin (`-K -`), never as arguments, so they never appear in the login node's process list.

Because every worker of an account shares `~/.ashley`, a worker Claude can hand work to another worker with `ashley send --worker <name>` from its shell; the daemon tells each worker who it is, which peers are alive and the exact command (`peerNote()` in `ashleyd.js`). A lead worker on the general container can therefore fan out to a GPU worker on a CUDA container and collect the result with `ashley result`.

The laptop side

- **Wizard** (`core/wizard.ts`): nine steps, each with `check()` and `run()`, over an abstract `WizardEnv` (settings, secrets, UI). The VS Code extension and the terminal CLI plug in their own UI. Rerunning does only what is missing.
- **SSH config** (`core/sshConfig.ts`): one managed block between markers, inserted first so it wins over any `Host *`. `MACs +hmac-sha1` because the grid's sshd offers nothing newer to Windows OpenSSH 9.5; `AddressFamily inet` because on Windows the IPv6 attempt to the login node times out first (13 s per connection measured, 2 s with IPv4 only); `IdentitiesOnly` so the agent's other keys are not tried; a wildcard `Host *.nas.grid.wayne.edu` with `ProxyJump` so no per-node edit is ever needed. On Windows the same block and key are mirrored into WSL.
- **Interactive steps** open a VS Code terminal whose *process* is `ssh` or `claude` (no shell, no quoting) and read its exit status.
- **Environment**: `environmentVariableCollection` (persistent) rather than editing `terminal.integrated.env` settings, so it can be cleared and does not leak into settings sync.
- **MCP**: `claude mcp add-json --scope user` with the server run by system `node` (>= 20) or by VS Code's own Electron with `ELECTRON_RUN_AS_NODE=1`, so students need not install Node. Fallback: `.mcp.json` in the workspace.

Xcode and other editors

`core/` never imports `vscode`. `dist/ashley-cli.js` runs the same wizard and the same grid calls from a terminal; `cli/ashley` is a ten-line ssh forwarder for machines without Node. The MCP server is editor-agnostic: any client that speaks MCP over stdio (Claude Code in Xcode's terminal, Claude Desktop, an Agent SDK script) can register it. A native Xcode extension is the wrong layer: Xcode Source Editor Extensions cannot open terminals, store secrets or run processes; the CLI plus MCP is the port.

What is deliberately not here

- No Nextflow yet. The queue and the worker are the substrate a pipeline can call (`ashley send` from a Nextflow process, `ashley result` in the next); the recipe's Nextflow

section stays the plan.
- No shared lab account, no cross-student pooling.
- No X11. Remote-SSH renders images; `srun --x11` remains available for the rare tool that needs a display.

What you can ask, and what you cannot

Written for students. “The grid Claude” is the Claude Code worker running in your Slurm job with your subscription; “the laptop Claude” is the Claude Code extension in your VS Code.

Things that work well

Ask the grid Claude to do work where the data is. It runs in your grid home with your environment: read a dataset’s manifest, write a measurement script, run it on ten records, look at the numbers, fix the script, run it on all 802, write the results file, draft the JOURNAL line. Say which folder (a session is bound to one), which files, and what “done” looks like.

Ask for a Slurm job, not a long foreground run. “Write `slurm/measure.sh` from the lab template, show it to me, and after I say submit, submit it and record the job id.” The worker can submit and poll jobs; it should not itself run a two-hour computation in one turn.

Ask it to continue. Sessions are named. “In session `thesis`, what did we decide about the exclusion list?” picks up where it stopped, days later, on any node.

Ask the laptop Claude to delegate. “Check the grid status, then ask the grid Claude to run the pelvic width demo on ten records and tell me the mean.” The laptop Claude uses the `ashley_*` tools; the heavy part happens on the grid; the answer returns to your panel.

Long tasks. `ashley_send` queues; `ashley_result` collects later. Close the laptop. The job runs until its wall time (24 h by default, 30 days maximum on the `primary` QoS).

Images. Open your grid files with **Ashley: Open my grid files on a compute node**; click a PNG. No X server.

Containers. Workers run the lab’s shared container once your lead has approved your access (the wizard’s last step files the request when everything else works; *Ashley: Lab access* shows its state); a lead can publish others (a CUDA one for training, an R/Bioconductor one) and you pick per worker. Two workers of yours can run different containers at once and hand work to each other.

GPUs. Start a worker with one GPU when the task trains or infers; the grid has H200, L40S, A30 and V100-class nodes. Ask for a GPU only for that worker; CPU workers start sooner.

Things that do not work, and why

You cannot	Because
Run the worker or VS Code's server on the login node	C&IT kills processes on `warrior`; Ashley only ever connects VS Code to a compute node and runs a half-second file poll on the login node
Use someone else's Claude account, or pool accounts across students	Anthropic's consumer terms: an account is for the person who holds it. Ashley rotates only *your* credentials (your subscription, then your own API key)
Get more than your plan's window	A Pro/Max window is per account. When it is exhausted the worker reports `rate_limited` with the reset time and waits; an API key with a spending cap is the only overflow
Put patient data, PHI, survey CSVs or contact lists on the grid	The lab's IRB umbrella (HPR 2026-269) covers public, de-identified datasets only; anything else needs its own filing first (RECIPE.md, data rules)
Push the grid Claude to `--dangerously-skip-permissions` for everything	It is available (`mode: bypassPermissions`) but the default `acceptEdits` lets it edit files and run tools while refusing destructive commands; use bypass only inside a project folder you can `git checkout` back
Ask for interactive programs (ITK-SNAP, a GUI) through the worker	The worker is headless. Run GUI tools locally on rsynced results, or with `srun --x11` in your own shell
Expect a reply in under a few seconds	A prompt reaches the worker in about 0.2 s over the open relay, but Claude Code's own start (about 5 s) comes before the first token; a real task takes minutes
Keep a worker forever	Jobs have wall limits; the `requeue` QoS may requeue the job (it resumes). Start another; queued requests wait
Store 200 GB in your home	Home is a 4 TB soft quota shared with everything else; Ashley's container and workspaces live on the lab's CephFS volume when you have access, and the doctor shows the quota

What the worker Claude can and cannot see

Can: your grid home and CephFS space, the internet (PubMed, Hugging Face, GitHub, Zenodo, pip), your GitHub and Hugging Face tokens (as environment variables, for `git push` and `hf download`), the Slurm commands, the lab's Python environments.

Cannot: your laptop's files (it is on the grid), your Claude password (it holds a token you can revoke), other students' homes, the login node's CPUs for anything heavy.

Good prompts

- "In session `pelvis`, working in `~/Student_Projects/projects/lee_pelvic_width`, read `proposal.md`, then write `code/measure_width.py` that measures the bi-iliac width on the label volumes listed in `lists/dev10.txt`, run it, and show me the ten numbers with units."
- "Write the Slurm script for the full 802 from the lab template. Do not submit; show it."
- "Compare the distribution in `results/widths.json` with the published values in `docs/normative.md`; flag anything outside the widest published SD."
- "Append today's entry to JOURNAL.md: what ran, job ids, what came out, what is next."

Prompts that will disappoint

- "Do my whole project." (No question, no data, no bar for failure. Start with PREREG.md.)
- "Make the number come out significant." (The recipe forbids editing results by hand and so does the worker's default mode; fix the script, rerun, report what you get.)
- "Download the CTs." (195 GB into a shared quota, on a request that may be repeated. Ask your lead for the shared path first.)

Installing and building

For a student (two minutes)

1. Install VS Code and, from the Extensions view, **Claude Code** (`anthropic.claude-code`) and **Remote - SSH** (`ms-vscode-remote.remote-ssh`). Sign in to Claude Code with your Claude Pro or Max account.
2. Get the file `ashley-wsu-1.0.0.vsix` from your lead (or the repository's `dist/` folder).
3. In VS Code: Extensions view, the `...` menu at the top right, **Install from VSIX...**, pick the file. Or from a terminal: `code --install-extension ashley-wsu-1.0.0.vsix`.
4. Command Palette (Ctrl/Cmd+Shift+P): **Ashley: Set up my research environment (wizard)**.

Before the wizard you need, once: a grid account (request it at tech.wayne.edu/kb/high-performance-computing naming your sponsor), and Google Authenticator set up through ondemand.grid.wayne.edu, Interactive Apps, 2FA Setup (VPN required for OnDemand).

What the wizard writes on your laptop

Where	What
`~/.ssh/id_ed25519_wsu`, `.pub`	a key pair without passphrase
`~/.ssh/config`	one block between `# >>> ashley` and `# <<< ashley` (a backup `config.ashley.bak` is kept)
OS keychain (Windows Credential Manager / macOS Keychain / libsecret)	GitHub, Hugging Face, Claude tokens
VS Code user settings `ashley.*`	your AccessID and preferences

On the grid it writes `~/.ashley/` (scripts, tokens with mode 600, workers, sessions), a block in `~/.bashrc`, and large files under `/rs/rs_grp_oschome/<AccessID>/ashley` or `~/ashley`.

To undo: remove the block from `~/.ssh/config`, delete the key, run **Ashley: Manage stored tokens** to delete tokens, uninstall the extension; on the grid, `rm -rf ~/.ashley` and the block in `~/.bashrc`.

Emulating a from-scratch install

`Ashley: Reset this laptop (emulate a fresh install)` forgets tokens, key and config block, settings, the Claude Code registration, and optionally Ashley on the grid (`ashley uninstall`, with `--purge-root` for the container). For a brand-new VS Code as well:

```
code --user-data-dir $env:TEMP\fresh-vscode --extensions-dir $env:TEMP\fresh-ext
```

install Claude Code, Remote - SSH and the Ashley VSIX in that window, and run the wizard. Delete the two folders afterwards.

Building from source

Requirements: Node.js 22 (the packaging tool refuses older), Git.

```
git clone <this repository> ashley
cd ashley/extension
npm install
npm run build          # bundles dist/extension.js, dist/mcp-server.js, dist/ashley-cli.js,
         copies ../grid into dist/grid
npm run test:unit      # 22 tests: ssh config, protocol, slurm parsing, and the daemon with a
         fake claude
npm run test:vscode    # downloads a VS Code build once, runs the integration suite
npm run package        # writes ../dist/ashley-wsu-<version>.vsix
code --install-extension ../dist/ashley-wsu-1.0.0.vsix
```

On a Windows machine without Node on `PATH`, the portable build works: download `node-v22.x-win-x64.zip` from nodejs.org, unzip, and prefix `PATH` with that folder for the commands above.

The extension bundles everything it needs (`vsce package --no-dependencies` plus esbuild), so the `.vsix` has no `node_modules`; students never run `npm`.

The terminal CLI (macOS, Linux, Xcode users)

`extension/dist/ashley-cli.js` is the wizard and the grid calls without VS Code:

```
node dist/ashley-cli.js setup            # the same nine steps, in the terminal
node dist/ashley-cli.js doctor
node dist/ashley-cli.js worker start --name default --hours 24
node dist/ashley-cli.js ask --session thesis "read PREREG.md and list its hypotheses"
```

`cli/ashley` (bash) and `cli/ashley.ps1` forward any command to the grid over ssh and need only an ssh client and the config block (`node dist/ashley-cli.js ssh-config` prints it).

For leads: the shared container library

One group-readable tree on the CephFS volume (`/rs/rs_grp_oschome/shared/ashley`) holds pre-built containers; students adopt them instead of building (no storage used, identical bits for everyone).

```
ssh wsu-grid "ashley library"                                    # what exists, which is the
        default
ssh wsu-grid "ashley library publish --name claude-node22 --image docker://node:22-bookworm --
        purpose 'general: Claude Code, Node 22, git, python3'"
ssh wsu-grid "ashley library publish --name claude-cuda --image docker://nvidia/cuda:12.4.1-
        runtime-ubuntu22.04 --gpu --purpose 'GPU: torch and inference'"
```

Each publish is a job (5 to 15 minutes) that builds the sandbox, installs the pinned Claude Code (the image must be able to run Node 22: Debian 12 / Ubuntu 22.04 or newer), merges the entry into `library.json` and makes the tree group-readable.

Who may read it is yours to decide, inside Ashley, with no HPC ticket. Run once:

```
ssh wsu-grid "ashley access init"
```

It makes `shared/` closed except the drop-box `shared/access/`. A student reaches **Join the lab**, the wizard's last step, only when everything else works and their Claude (Pro at the minimum), GitHub and Hugging Face tokens verify; the step files one request with a note. **Ashley: Lab access to the shared library** shows you the queue (pending first) with Approve, Deny, Deny with message, Remove, Reset and a direct Grant; the Ashley view shows the pending count. Approve is a POSIX ACL over the whole lab folder, instantly effective and revocable; the student's next worker uses the lab container without building anything. The command line equivalents:

```
ssh wsu-grid "ashley access admin list"
ssh wsu-grid "ashley access admin approve ab1234"
ssh wsu-grid "ashley access admin deny ab1234 --message 'finish the onboarding first'"
ssh wsu-grid "ashley access admin reset ab1234"      # lets them ask again
```

Updating

Install the new `.vsix` the same way; run **Ashley: Install or update Ashley on the grid** to push new grid scripts (a job rebuilds the container only if the pinned Claude Code version changed).

Troubleshooting

Run **Ashley: Check everything (doctor)** first. It prints one line per check on the laptop and, when the connection works, the grid's own checks.

Symptom	Cause	Fix
`no matching MAC found. Their offer: hmac-sha1`	your `ssh` config lacks the Ashley block	rerun the wizard (step 3 rewrites the block with `MACs +hmac-sha1`)
Every grid action takes 13 s	IPv6 attempt on Windows	the block sets `AddressFamily inet`; rerun the wizard if you edited the file
Step 4 rejects the password or code three times	wrong password/code, an account not yet created, or Authenticator not set up	check ondemand.grid.wayne.edu logs you in; then rerun the wizard. If the in-panel login cannot reach the grid at all, the wizard falls back to a terminal for the same two prompts
Step 5: GitHub sign-in does not open a browser	VS Code's GitHub sign-in is disabled or blocked	the panel shows a pasted-token field with the token page pre-filled
Step 6/7: the panel does not notice the copied token	clipboard access blocked, or the copy did not happen	paste it into the field; the check is the same
`Permission denied (publickey...)` after step 4	the key was not appended to `authorized_keys`	rerun step 4; on the grid, `cat ~/.ssh/authorized_keys` should contain "ashley"
Ashley view: *Grid installing* for more than 30 min	the bootstrap job is queued	`ssh wsu-grid squeue -u $USER`; if it pends for long, `ssh wsu-grid "~/.ashley/bin/ashley bootstrap --resubmit --partition secp --qos secondary"`
Worker shows PENDING for long	the `requeue` QoS pool is busy	start it with the `secp` / `secondary` settings (`ashley.worker.partition`, `ashley.worker.qos`)
Ask returns `no enabled Claude account`	step 7 was skipped	**Ashley: Store or replace my Claude token for the grid**
Ask returns `Failed to authenticate`	the token expired or was revoked	run `claude setup-token` again and store the new token
Ask returns `rate_limited` with a time	your subscription window is exhausted	wait for the time shown, or add an API key (step 8)
Remote-SSH: *Connecting with SSH timed out*	the compute node's sshd needs about 25 s per session; the first	Ashley sets `remote.SSH.connectTimeout` to 300 when it opens a node; if you

Symptom	Cause	Fix
	connection also downloads the server	connect by hand, set it yourself in Settings and connect again
Remote-SSH window: *You are connecting to an OS version that is unsupported by Visual Studio Code*	the grid is CentOS 7; VS Code's FAQ says the banner and a dialog appear on every such connection even when the sysroot makes it work	dismiss it, or *Don't show again*; nothing else to do
Remote-SSH: *glibc* or *exit code 207*	the C&IT sysroot variables are not in `~/.bashrc`, or a `return` guard precedes them	`ssh wsu-grid "~/.ashley/bin/ashley install"` rewrites the block; move it above any `[ -z "$PS1" ] && return`
VS Code connects to `warrior` instead of a node	a job was not running	start a worker; Ashley connects only to nodes where you hold a job
`home quota` above 90 %	your home is full	large files belong on the lab volume; ask your lead for the path; `ashley.gridRoot`
The MCP tools do not appear in Claude Code	the bridge was not registered, or Claude Code was open before registration	**Ashley: Connect the Claude Code extension to the grid**, then start a new conversation

Logs

- Laptop: *View, Output, Ashley* (every command and its exit code).
- Grid: `~/.ashley/log/bootstrap_<job>.out`, `~/.ashley/workers/<name>/log/ashleyd.log`, `~/.ashley/workers/<name>/log/slurm_<job>.out`, and every request's `outbox/<id>.events.jsonl`.
- Remote-SSH: *View, Output, Remote - SSH*.

Undo

Laptop: remove the block from `~/.ssh/config` (a `config.ashley.bak` is kept), delete `~/.ssh/id_ed25519_wsu*`, **Ashley: Manage stored tokens** to delete tokens, uninstall the extension. Grid: `ashley worker stop`, `rm -rf ~/.ashley`, remove the block from `~/.bashrc`, delete the container folder under your storage root.

Appendix

- `docs/RESEARCH.md`: the research behind the design (Claude Code headless mode, subscription limits and terms, containers on this grid, multi-account tools, orchestration patterns, VS Code Remote-SSH on old glibc, Xcode, literature APIs), with sources and which findings were independently verified.
- `docs/CHECKPOINT_2026-09-28.md`: the state of the project on the day this manual was built, including what was tested and what was not.
- `Student_Projects/RECIPE.md`: the lab's method; Ashley is the on-ramp to it.
- C&IT: "How to SSH to the Grid in Visual Studio Code" and "How do I set up and use Google Authenticator for the Grid" (services.wayne.edu knowledge base).